

What is a Network?

A **network** is created when two or more computing devices (like computers, laptops, servers, printers, or smartphones) are connected so they can communicate and share resources.

The connection can happen through:

- **Wired media** (Ethernet cables, fiber optic cables)
- **Wireless media** (Wi-Fi, Bluetooth, cellular networks)

At its core, a network allows devices to:

- Exchange data (messages, files, emails)
- Share hardware (printers, scanners)
- Access shared storage (file servers, cloud drives)
- Use shared services (databases, applications)

The **Internet** is the largest example of a network. In fact, it is often described as a “*network of networks*” because it connects millions of smaller networks worldwide.

Types of Networks

Networks are classified based on geographical coverage.

LAN – Local Area Network

A **LAN** connects devices within a small area such as:

- Homes
- Offices
- Schools
- Buildings

Examples:

- Wi-Fi network in your house
- Ethernet-connected computers in an office

LANs are typically fast and privately managed.

MAN – Metropolitan Area Network

A **MAN** covers a larger area — usually a city.

Example:

- A city-wide network connecting multiple office branches
- Cable TV network infrastructure across a city

WAN – Wide Area Network

A **WAN** connects networks over large geographical areas — countries or continents.

Example:

- The Internet
- Corporate networks connecting global branches

WANs often use technologies like:

- Fiber optic cables
- Satellite links
- Submarine cables

SONET – Synchronous Optical Network

SONET is a high-speed fiber optic standard used mainly for long-distance communication, including undersea cable systems.

It enables reliable, high-bandwidth data transmission across continents.

Network Devices (Core Components)

A network cannot function without hardware components that control and manage communication.

Switch

A **switch** connects multiple devices within the same network (usually a LAN).

- It forwards data only to the intended device.

- It uses **MAC addresses** to identify devices.

Switches are intelligent and reduce unnecessary traffic.

Router

A **router** connects different networks together.

Example:

- Your home router connects your LAN to the Internet.

It routes data based on **IP addresses**.

Modem

A **modem** converts digital signals into analog signals (and vice versa).

- Used to connect your network to an Internet Service Provider (ISP).
- The word comes from **Modulator + Demodulator**.

Hub (Legacy Device)

A **hub** connects multiple devices but broadcasts data to all devices.

- Not intelligent
- Causes unnecessary traffic
- Largely replaced by switches

NIC – Network Interface Card

The **NIC** allows a device to connect to a network.

- Can be wired (Ethernet port)
- Can be wireless (Wi-Fi adapter)
- Has a unique **MAC address**

Modern motherboards come with built-in NICs.

Bridge

A **bridge** connects multiple LAN segments.

- It reduces broadcast traffic.
- Works using MAC addresses.
- Mostly replaced by modern switches.

What is a Protocol?

A **protocol** is a set of rules that defines how data is transmitted between devices.

Without protocols, devices wouldn't know:

- How to send data
- How to format data
- How to acknowledge receipt
- How to handle errors

Common Network Protocols:

- **HTTP** – Transfers web pages
- **HTTPS** – Secure version of HTTP
- **TCP** – Reliable data transmission
- **IP** – Logical addressing and routing
- **FTP** – File transfer
- **SMTP** – Email sending

Protocols ensure structured and standardized communication.

What is an IP Address?

An **IP (Internet Protocol) address** is a unique logical address assigned to a device in a network.

Think of it as a digital home address.

It helps devices:

- Identify each other
- Locate each other

- Communicate across networks

Types of IP Addresses

IPv4

- 32-bit address
- Written in four decimal numbers (octets)
- Example: 192.168.1.1
- Total addresses: ~4.3 billion

Each octet = 8 bits

4 octets × 8 bits = 32 bits

Due to limited address space, IPv4 is gradually being replaced.

IPv6

- 128-bit address
- Written in hexadecimal
- Example: 2001:0db8:85a3:0000:0000:8a2e:0370:7334

Benefits:

- Massive address space
- Better security
- Designed for future scalability

Public vs Private IP

Public IP

- Assigned by ISP
- Accessible over the Internet
- Globally unique

Private IP

Used inside local networks.

Common ranges:

- 192.168.0.0 – 192.168.255.255
- 10.0.0.0 – 10.255.255.255
- 172.16.0.0 – 172.31.255.255

Private IPs cannot be accessed directly from the Internet.

Static vs Dynamic IP

Static IP

- Manually assigned
- Does not change
- Used for servers, printers, hosting

Dynamic IP

- Assigned automatically by **DHCP**
- Changes periodically
- Used in home networks

IPv4 Address Classes (Classful System – Historical)

IPv4 was originally divided into classes based on the first octet.

Class	Range	Purpose
A	1–126	Large networks
B	128–191	Medium networks
C	192–223	Small networks
D	224–239	Multicast

E	240–255	Reserved
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Today, modern networking uses **CIDR (Classless Inter-Domain Routing)** instead of traditional classes for better flexibility.

A network is not just about connecting devices.
It's about enabling structured, reliable, and secure communication.

From:

- Physical cables
To:
- IP addressing
To:
- Protocol rules
To:
- Global Internet infrastructure

Networking is the foundation of modern digital communication.

Important Notes About Special IPv4 Addresses

Why 0.0.0.0 is Not a Class A Address

Although Class A officially starts from **1.0.0.0 to 126.0.0.0**, the address **0.0.0.0** is excluded.

0.0.0.0 is a special-purpose address.

It is used to represent:

- “Unknown address”
- “This network”
- Default route in routing tables

It is **not assigned to devices** and is not usable as a normal host IP.

127.x.x.x – Loopback Range

The entire range:

127.0.0.0 to 127.255.255.255

is reserved for **loopback addresses**.

The most commonly used address is:

127.0.0.1 → localhost

What is Loopback?

A **loopback address** allows a device to communicate with itself.

When you ping 127.0.0.1:

- The data never leaves your computer.
- It stays inside the operating system.
- It tests whether the TCP/IP stack is working correctly.

Why is it useful?

- Testing web servers locally
- Testing network applications
- Diagnosing network configuration issues
- Verifying that the network stack is functioning

Any IP in the **127.x.x.x** range loops back to the same machine.

IP Address Structure: Network ID & Host ID

An IPv4 address has two logical parts:

1. **Network ID** → Identifies the network
2. **Host ID** → Identifies the specific device inside that network

Any device assigned an IP address is called a **host**.

Network ID in Classful Addressing

In the older class-based system:

Class	Network Portion	Host Portion
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Class A	1st Octet	Last 3 Octets
Class B	1st & 2nd Octets	Last 2 Octets
Class C	1st, 2nd & 3rd Octets	Last Octet

Example (Class C):
IP: **192.168.1.10**

- Network ID → 192.168.1
- Host ID → 10

Device Connection Rules

Same Network ID → Direct Communication

If two devices share the same Network ID, they can communicate directly without a router.

Example:

Device A: 17.0.0.1
Device B: 17.0.4.2

Class A → Network ID = 17

Both devices belong to network 17.

They can communicate directly.

Different Network IDs → Router Required

Example:

Device A: 17.0.0.1
Device B: 192.168.1.5

Network IDs:

- Device A → 17
- Device B → 192.168

These are different networks.

A **router is required** to forward traffic between them.

Subnetting – Breaking a Network into Smaller Networks

Subnetting means dividing a large network into smaller, manageable networks.

Why do we do this?

- Better performance
- Better security
- Better IP management
- Smaller broadcast domains

Example: 192.168.1.0/24

The “/24” is CIDR notation.

It means:

- 24 bits are used for the Network
- 8 bits are used for Hosts
- Subnet mask = 255.255.255.0

Total IPs:

$2^8 = 256$ addresses

Range:

192.168.1.0 → 192.168.1.255

Usable host addresses:

254

(First is Network ID, Last is Broadcast)

Dividing /24 into Two Subnets

If we change the subnet to **/25**:

- 25 bits → Network
- 7 bits → Host
- Subnet mask = 255.255.255.128

$2^7 = 128$ addresses per subnet

Subnet 1: 192.168.1.0/25

Range:

192.168.1.0 → 192.168.1.127

- Network Address → 192.168.1.0
- Broadcast Address → 192.168.1.127
- Usable Hosts → 126

Subnet 2: 192.168.1.128/25

Range:

192.168.1.128 → 192.168.1.255

- Network Address → 192.168.1.128
- Broadcast Address → 192.168.1.255
- Usable Hosts → 126

Why Subnetting is Powerful

Improves Performance

Reduces broadcast traffic.

Each subnet becomes its own broadcast domain.

Enhances Security

Different departments can be isolated:

- HR
- Finance
- IT

- Guest Network

Each can be placed in separate subnets.

Efficient IP Utilization

Prevents wasting IP addresses.

You allocate only what each department needs.

CIDR – Classless Inter-Domain Routing

CIDR replaced the old classful system.

Instead of fixed classes (A, B, C), CIDR allows flexible allocation using prefix notation.

Example:

- /8
- /16
- /24
- /30

CIDR improves:

- IP address conservation
- Routing efficiency
- Internet scalability

Common CIDR Prefixes

CIDR	Subnet Mask	Total IPs	Usable Hosts
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/29	255.255.255.248	8	6
/30	255.255.255.252	4	2

CIDR gives modern networks flexibility that the class-based system never could.

Networking is structured logic.

- IP Address → Identifies device
- Network ID → Identifies group
- Router → Connects groups
- Subnetting → Organizes groups
- CIDR → Makes allocation flexible

From simple LAN communication to global Internet routing, everything works because these logical rules are applied consistently.

Network Models

To understand how devices communicate over a network, we use **network models**. These models act like structured blueprints that describe how data moves from one device to another.

There are two main models:

1. **OSI Reference Model**
2. **TCP/IP Model**

OSI Reference Model (7 Layers)

The **OSI (Open Systems Interconnection) Model** was developed by the **ISO (International Organization for Standardization)** to standardize network communication.

It divides communication into **7 logical layers**, where each layer has a specific responsibility.

Think of it like sending a package:

- One layer prepares the message
- One encrypts it
- One adds addressing
- One physically sends it

Each layer performs a role and passes the data to the next layer.

The 7 Layers of OSI (Top to Bottom)

7 Application Layer

The layer closest to the user.

It provides network services directly to applications.

Examples:

- HTTP
- FTP
- SMTP
- DNS

This is where web browsing, email, and file transfers begin.

6 Presentation Layer

Responsible for:

- Data translation
- Encryption / Decryption
- Compression

Example:

- SSL/TLS encryption
- Image formats like JPEG, PNG
- MIME (email attachments)

It ensures both systems understand the data format.

5 Session Layer

Manages sessions between two systems.

It:

- Establishes connections
- Maintains sessions
- Terminates sessions

Example:

- NetBIOS
- PPTP (VPN tunneling)

4 Transport Layer

Ensures reliable or fast data delivery.

Protocols:

- TCP (Reliable)
- UDP (Fast but unreliable)
- SCTP

It controls:

- Flow control
- Error detection
- Retransmission

This layer uses **port numbers**.

3 Network Layer

Handles:

- Logical addressing
- Routing decisions
- Packet forwarding

Protocols:

- IP (IPv4/IPv6)
- ICMP (ping)
- Routing protocols

Routers operate at this layer.

2 Data Link Layer

Responsible for:

- MAC addressing
- Error detection (frames)
- Communication between directly connected devices

Examples:

- Ethernet
- ARP
- PPP
- HDLC

Switches operate at this layer.

1 Physical Layer

The lowest layer.

It deals with:

- Electrical signals
- Cables
- Voltages
- Bit transmission

Examples:

- Ethernet cabling
- Fiber optics
- USB

Port Numbers and Protocols

Port numbers help deliver data to the correct application.

Some common ports:

Protocol	Port	Purpose
HTTP	80	Web browsing
HTTPS	443	Secure web

FTP	20, 21	File transfer
SMTP	25	Sending email
POP3	110	Receiving email
IMAP	143	Receiving email
DNS	53	Name resolution
SSH	22	Secure remote access

Ports work mainly at the **Transport Layer (TCP/UDP)** and are used by Application Layer protocols.

TCP/IP Model (Real-World Model)

The TCP/IP model is the practical model used on the Internet.

It has **4 layers**, not 7.

TCP/IP Layers:

1. Application Layer
2. Transport Layer
3. Internet Layer
4. Network Interface Layer

Mapping OSI to TCP/IP

TCP/IP	Equivalent OSI Layers
Application	Application + Presentation + Session
Transport	Transport
Internet	Network
Network Interface	Data Link + Physical

The TCP/IP model is simpler and used in real-world networking.

Important Protocols Explained Clearly

HTTP (Hypertext Transfer Protocol)

HTTP is:

- Client-server based
- Stateless (server does not remember previous requests)

Client sends → HTTP Request

Server sends → HTTP Response

HTTP Status Code Categories

Range	Meaning
200s	Success
300s	Redirection
400s	Client error
500s	Server error

Examples:

- 200 → OK
- 404 → Not Found
- 500 → Internal Server Error

How Login Works if HTTP is Stateless?

If HTTP does not remember users, how does Amazon keep you logged in?

The answer: **Cookies**

Cookies:

- Small data stored in your browser
- Sent with every HTTP request

- Used for authentication and session management

They allow servers to recognize returning users.

Email Protocols

SMTP

Used for sending email between mail servers.

POP3

Downloads email from server to client.

IMAP

Keeps email synchronized on server and client.

FTP (File Transfer Protocol)

Used to:

- Upload files
- Download files
- Transfer files between hosts

SSH (Secure Shell)

Used by system administrators to:

- Remotely access systems
- Securely manage servers
- Encrypt remote sessions

Port: 22

TCP vs UDP

TCP

- Connection-oriented
- Reliable
- Uses acknowledgments
- Retransmits lost packets

Used when accuracy matters (websites, databases).

TCP Three-Way Handshake

1. SYN
2. SYN-ACK
3. ACK

This ensures a stable connection before data transfer.

UDP

- Connectionless
- No acknowledgments
- Faster
- No guarantee of delivery

Used for:

- Video streaming
- Online gaming
- DNS queries

Speed over reliability.

Routing – How Packets Travel Between Networks

Routing decides how packets move from one network to another.

Routers use **routing tables**.

When deciding a route:

- More specific routes are checked first.
- If no specific match is found → default route (0.0.0.0/0) is used.
- Default route usually forwards traffic to the Internet.

Routing ensures global connectivity.

DNS (Domain Name System)

DNS translates domain names into IP addresses.

Example:

www.example.com → 93.184.216.34

Computers understand IP addresses, humans prefer names.

How DNS Works (Step-by-Step)

1. You type a website name.
2. Browser checks cache.
3. If not found, query goes to DNS resolver.
4. Resolver contacts:
 - Root Server
 - TLD Server (.com, .org)
 - Authoritative Server
5. IP address is returned.
6. Browser connects to server.

Domains, Zones, and Delegation

Domains

Structured like a tree:

Root → .com → example.com → sub.example.com

Zones

A portion of DNS managed by an organization.

Example:

.com managed by registry operator.

Delegation

Control of a subdomain can be handed over using **NS records**.

Example:

Control of example.com delegated to a company's nameservers.

Common DNS Record Types

Record	Purpose
A	Maps domain to IPv4
AAAA	Maps domain to IPv6
CNAME	Alias
MX	Mail server
NS	Nameserver
TXT	Text verification
PTR	Reverse lookup

Each record has a **TTL (Time To Live)**.

DHCP (Dynamic Host Configuration Protocol)

DHCP automatically assigns:

- IP address
- Subnet mask
- Default gateway
- DNS server

When you connect to Wi-Fi, DHCP configures your device automatically.

Network Components & Services

Routers

Connect different networks and forward packets.

Switches

Connect devices inside the same network using MAC addresses.

Firewalls

Control traffic based on security rules.

Load Balancers

Distribute traffic across multiple servers.

VPN

Creates a secure encrypted tunnel over the Internet for remote access.

Networking works in layers.

From:

- Physical cables
To:
- MAC addressing
To:
- IP routing
To:
- Port-based communication
To:
- Application services

Everything follows structured rules.

Understanding OSI and TCP/IP models helps you:

- Troubleshoot efficiently
- Design networks properly
- Understand cloud architecture
- Prepare for networking interviews

Network Troubleshooting Tools:

1. ping

Purpose: Test internet network connections.

How It Works: Uses the **ICMP ECHO_REQUEST** to get an **ICMP ECHO_RESPONSE** from a remote host.

Usage: For basic troubleshooting, you can run `ping www.google.com` to check network connectivity and see response times and packet loss.

2. traceroute (or tracert on Windows)

Purpose: Track the route packets take to reach their destination.

How It Works: Sends **UDP probes** with increasing TTL values, showing each router along the route and the delay in reaching it.

Usage: Helps find which gateway is causing a delay by showing response times and where packets fail (indicated by `*`).

3. telnet

Purpose: Test network connections and protocols.

How It Works: Attempts to establish a connection to a specified IP and port.

Usage: Test if a specific service is reachable, e.g., `telnet google.com 443`.

4. curl

- ♦ **Purpose:** Transfer data using multiple protocols, often for HTTP requests.

- ♦ **Usage:**

- Basic GET request: `curl http://example.com`.

- Check headers: `curl -I http://example.com`.

- POST request: `curl -X POST http://example.com.`
- Save response to file: `curl http://example.com/file -o output.file`

5. dig (Domain Information Groper)

- ♦ **Purpose:** Troubleshoot **DNS** problems and verify DNS records.
- ♦ **How It Works:** Performs DNS lookups and displays information such as IP addresses.
- ♦ **Usage:** `dig google.com` to get information like IP addresses, TTL, and DNS record types.

6. netstat

Purpose: Show network connections and port listening information.

Usage:

- `netstat -lp`
: List listening servers and their program names. `netstat -a` :
Show all active ports.
- `netstat -r` : Show routing table.

7. nmap (Network Mapper)

Purpose: Discover hosts and services on a network.

How It Works: Sends raw packets to identify hosts, services, and operating systems.

Usage:

- Discover hosts: `nmap -sn 172.31.44.35/20`
- Scan ports on a host: `nmap -A 172.31.36.237`

8. ssh (Secure Socket Shell)

- ♦ **Purpose:** Securely connect to remote machines to execute commands.
- ♦ **Usage:**
 - `ssh username@hostname` .
 - Secure and encrypted, used for remote management and file transfers.

9. scp (Secure Copy Protocol)

Purpose: Securely copy files between local and remote hosts.

Usage:

- Copy file to a remote server: `scp localfile.txt user@remote:/path/to/destination .`

These tools are invaluable for **network diagnostics, troubleshooting, and secure communications**, which are critical skills for any **DevOps engineer**.